

DAI 1000

Digital and AI Literacy

**L2 - Information Technology
Development**

Objectives

By the end of this class, you should be able to:

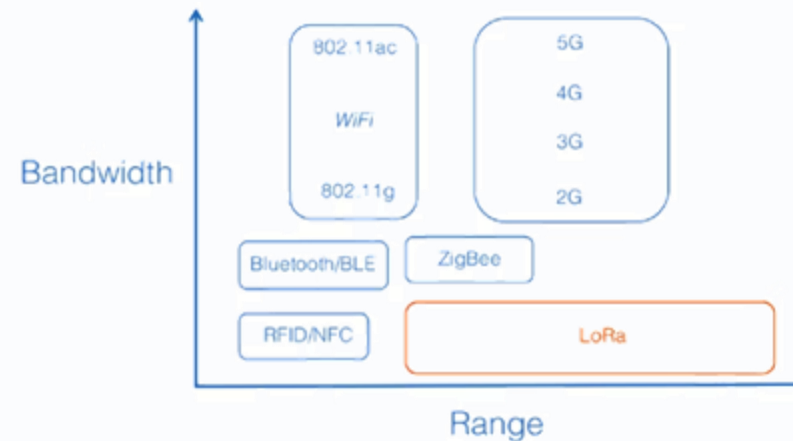
- have basic understanding of key information technologies (IT)
- recognize the foundational role of IT infrastructure in improving daily life
- lay the groundwork for understanding modern AI and IT ecosystems

Introduction to Information Technology (IT)

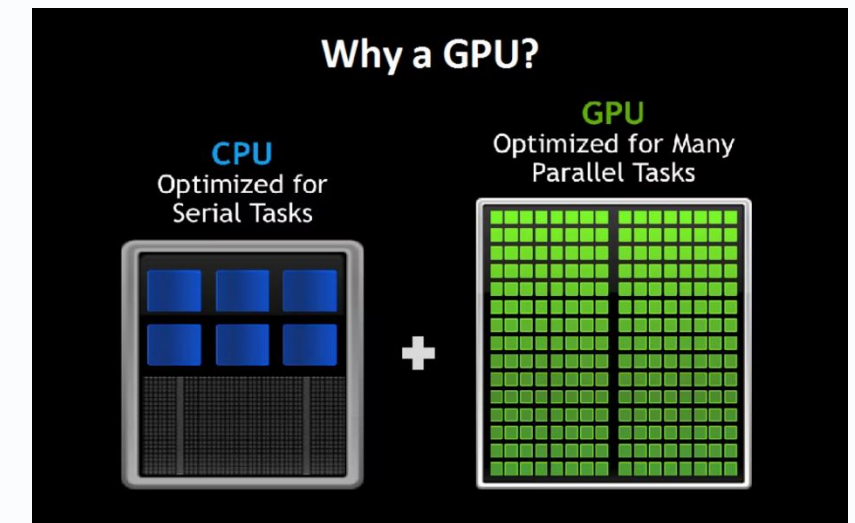
The use of computers, communication systems, and other devices to process, store/retrieve, manipulate, and transmit information.



process



transmit



manipulate

Interdependency of Technologies

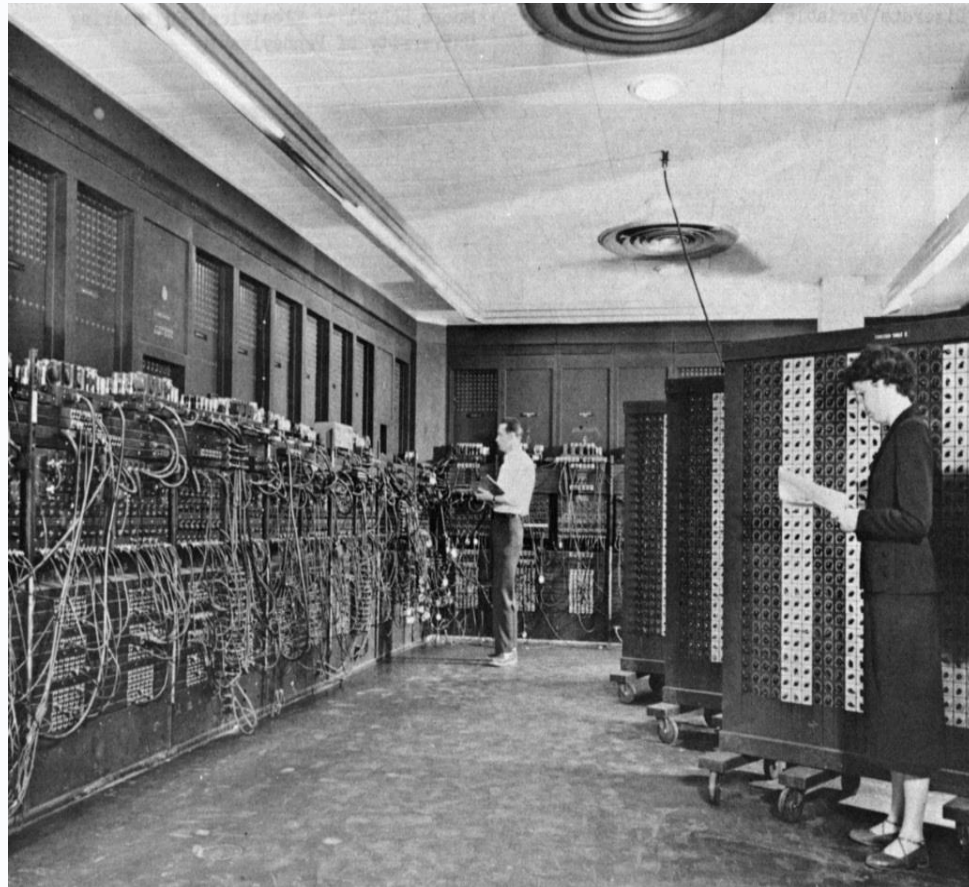
IT developments include the evolution of key technologies

- Computing Evolution and Chips → hardware basis
 - Sensors and IoT data generation
 - Internet and Connectivity data generation & transmission
 - Big Data ←
 - Cloud Computing where to process data
 - Smart City x-smart applications of cities
- 

Content

- Computing Evolution and Chips
- Internet and Connectivity
- Cloud Computing

The Computing Evolution



ENIAC (1946, University of Pennsylvania)
17,468 vacuum tubes, 170 m²
30 tons, 5000 operations per second

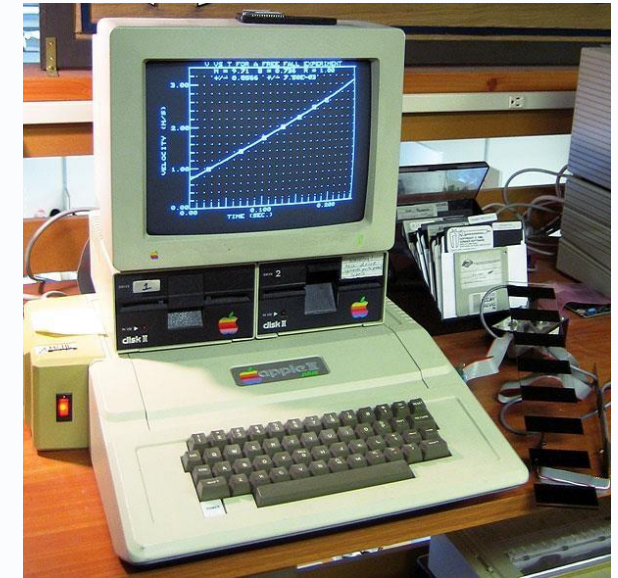
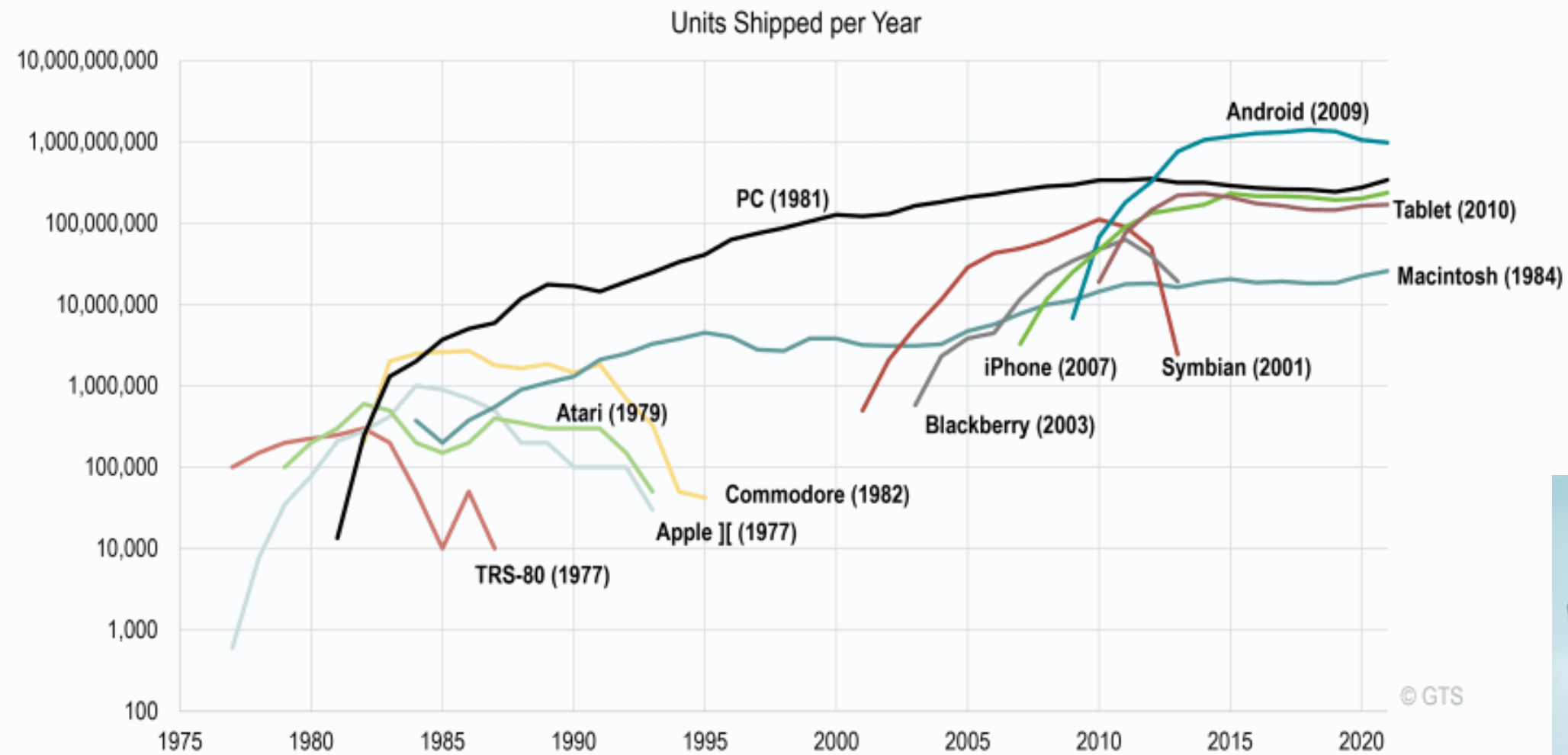


Personal computing
From room-sized giants
to palm-sized power

Personal Computing



Personal Computing



Super Computing

Tianhe-2 (天河二号):

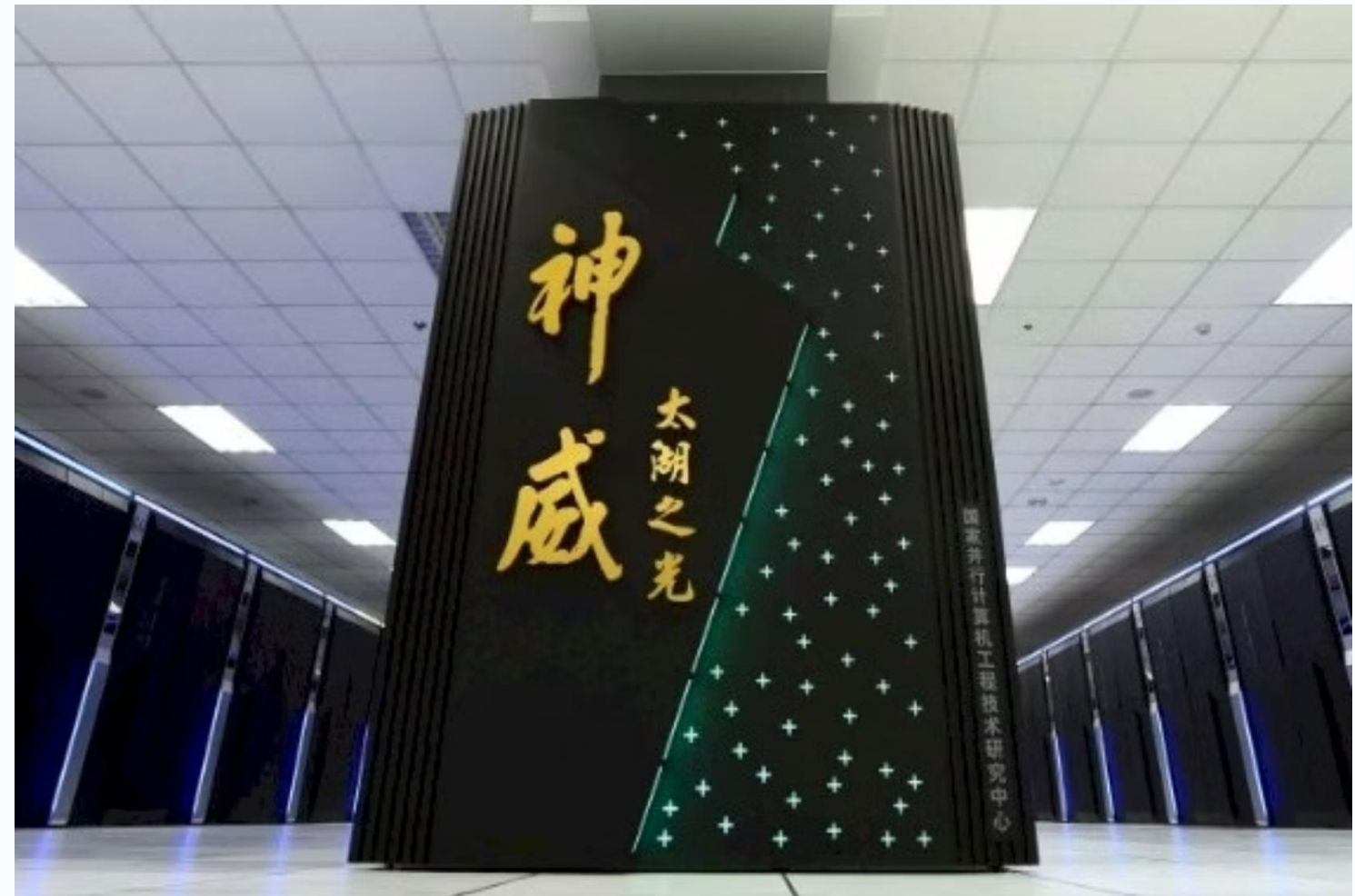
- developed by a team of 1,300 scientists and engineers;
- was the world's fastest supercomputer from 2013 until 2016



Super Computing

The Sunway TaihuLight (神威)

- was the fastest supercomputer from 2016 until 2018;
- 93 TeraFlops (1012 floating point operations per second)



Super Computing

Country	System	Manufacturer	Hardware supplier	Max Performance (PFlop/s)	Power (kW)
 US	El Capitan	HP Enterprise	AMD	1,742	29,581
 US	Frontier	HP Enterprise	AMD	1,353	24,607
 US	Aurora	HP Enterprise	Intel	1,012	38,698
 US	Eagle	Microsoft	Intel NVIDIA	561	
 Italy	HPC6	HP Enterprise	AMD	478	8,461
 Japan	Supercomputer Fugaku	Fujitsu	Fujitsu	442	29,899
 Switzerland	Alps	HP Enterprise	NVIDIA	435	7,124
 Finland	LUMI	HP Enterprise	AMD	380	7,107
 Italy	Leonardo	EVIDEN	Intel NVIDIA	241	7,494
 US	Tuolumne	HP Enterprise	AMD	208	3,387

Super Computing

Currently, the fastest one is:

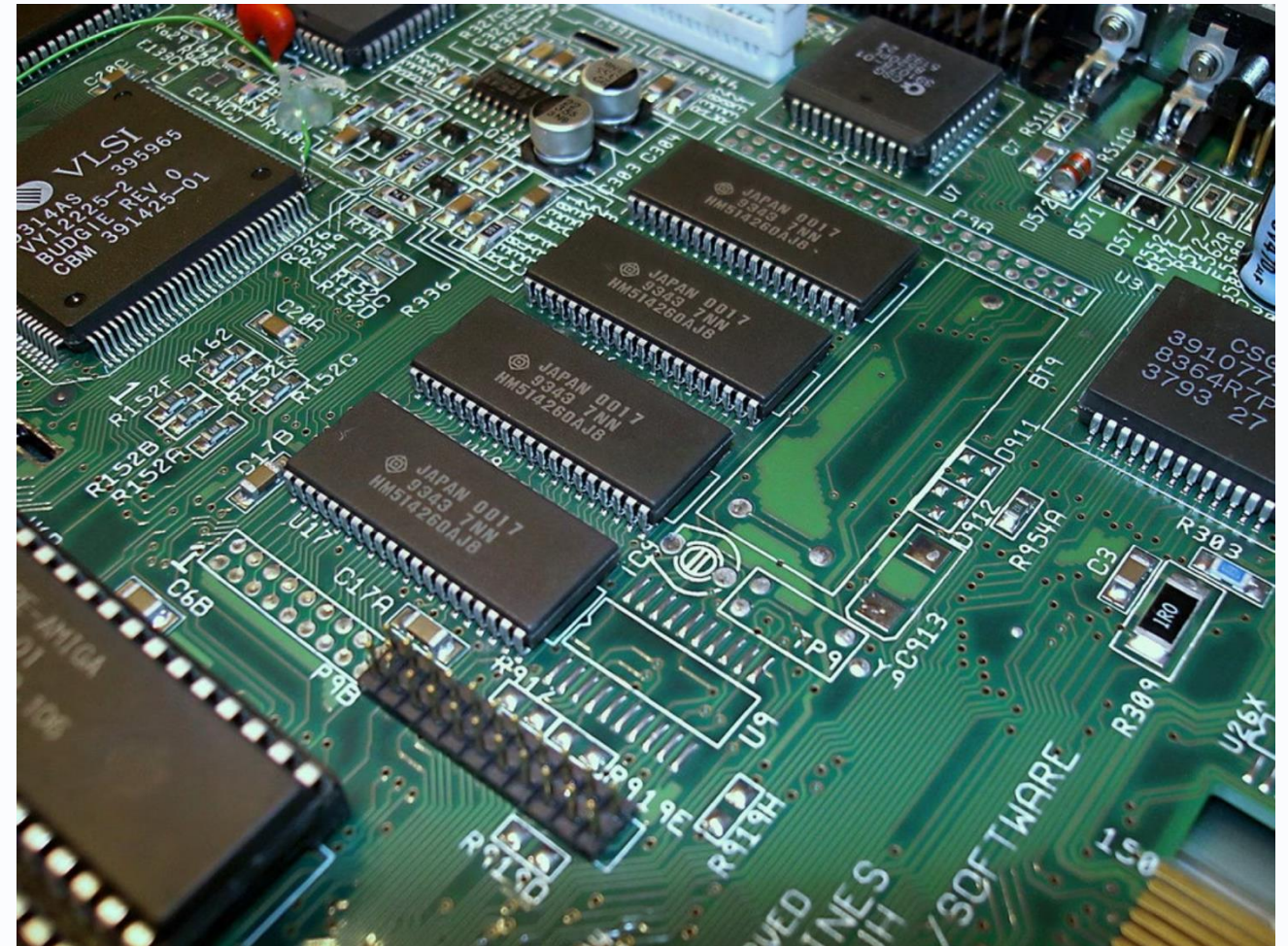
El Capitan

https://www.bilibili.com/video/BV1oAa8zyEZR/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f



Chips

- How are chips made?
- Where are chips used?
- Who makes the chips?
- What caused the shortages?
- Consequences of the shortages



Chips

How Are Microchips Made?

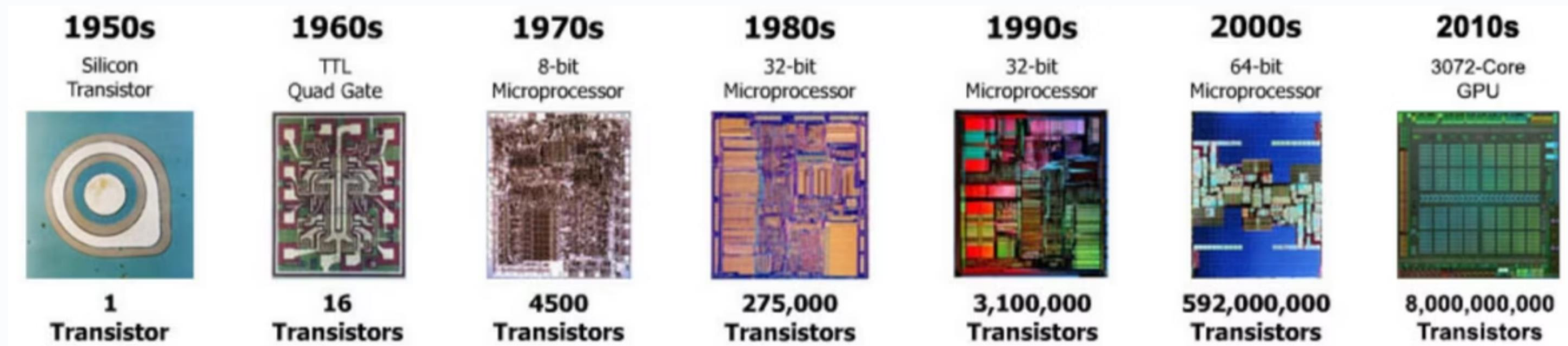
https://www.bilibili.com/video/BV1r9XzY9Eid/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

Scan QR Code to Answer Questions

Moore's Law: Growth in computing power made it possible to train AI models

Transistor density on integrated circuits doubles every two years

- Proposed by Gordon E. Moore (Intel co-founder) in 1965
- Doubling performance/capacity every two years?
- Halving price every two years?



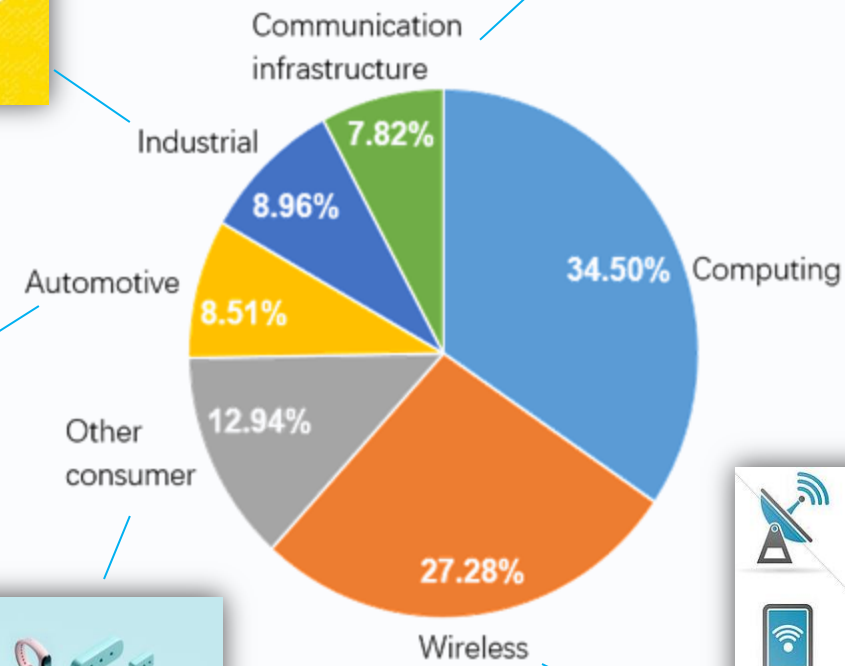
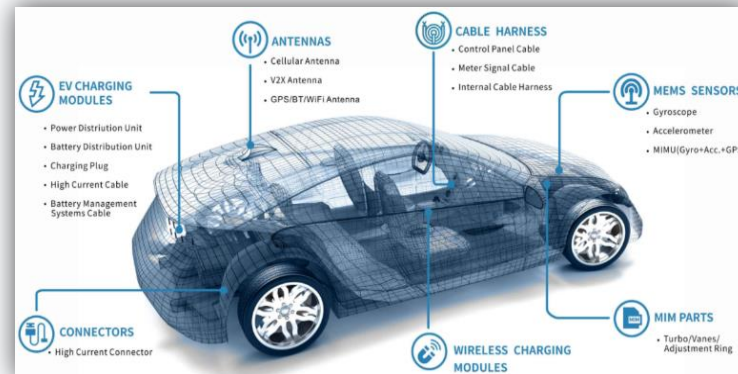
Moore's Law is Slowing!

Why is my PC not as fast as predicted by the Law?

- Doubling chip density now takes longer, rather than two years
- Technologies for keeping up with Moore's Law are getting more difficult to implement (low ROI)
- Andy and Bill's Law: What Andy gives, Bill takes away. In other words, complex software soaks up the new chip's power



Where are chips used ?



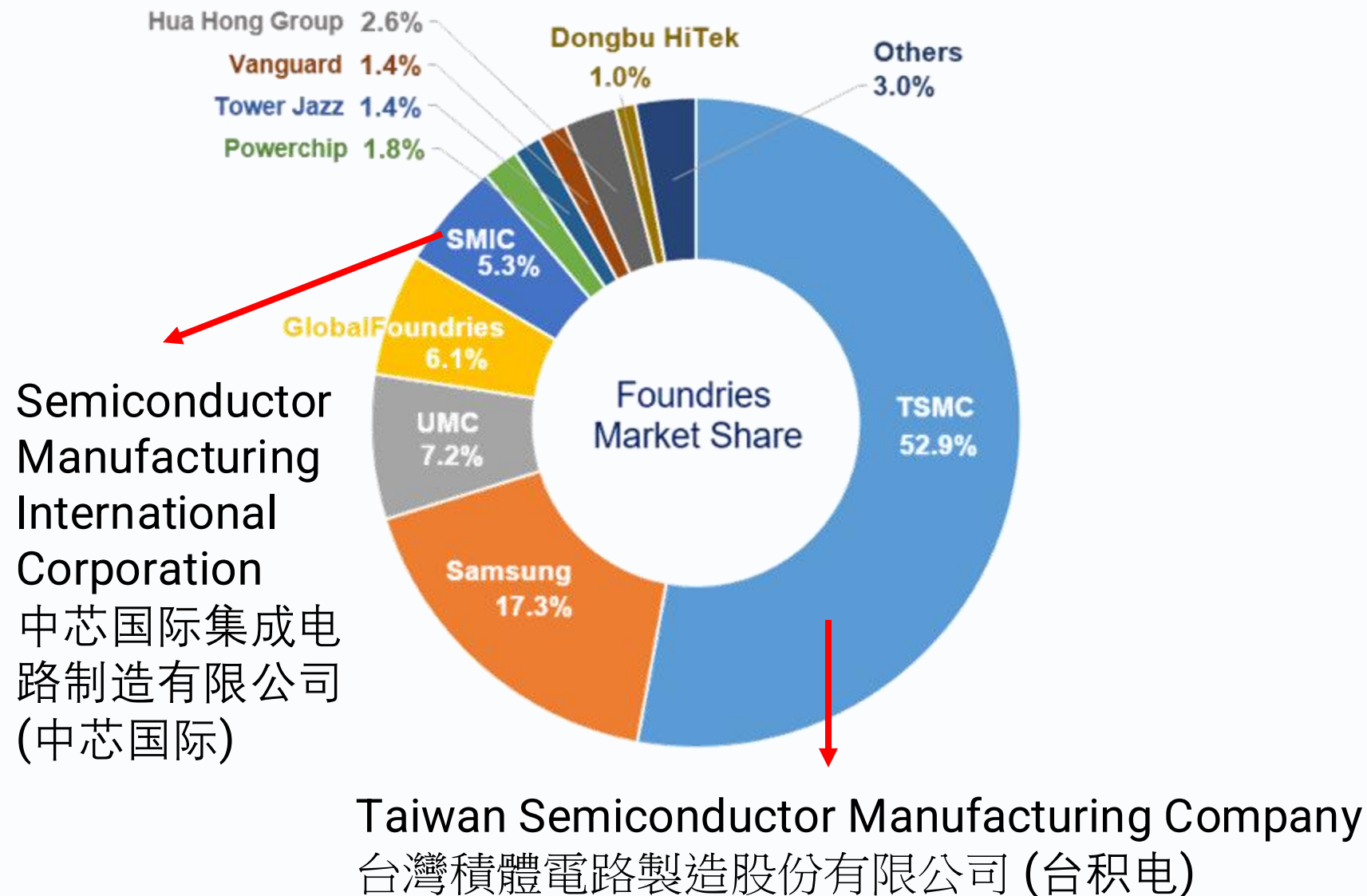
<https://spectrum.ieee.org/chip-shortage>

<https://www.n-redc.co.jp/en/applications/industrial-approach>

Who makes the chips ?

Foundries (or fabs) are the companies that fabricate chips

Chips come out of fabs, go through the supply chain, then used in various products



Inside one of TSMC fabs

<https://www.trendforce.com/presscenter/news/20210831-10914.html>

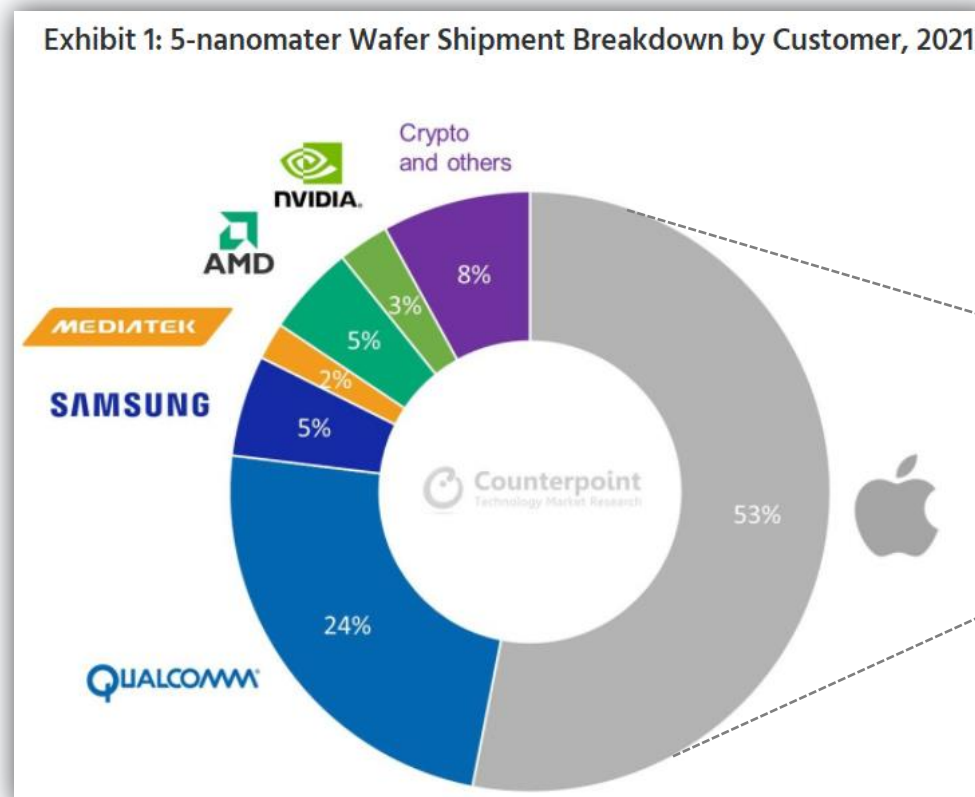
<https://www.tsmc.com/english/dedicatedFoundry/manufacturing/engineering>

Who makes the chips ?

- Chips consist of transistors
e.g. A16 Bionic integrates 16 billion transistors
- Transistor size is getting smaller, described by “technology node”.
e.g. A16 was fabricated with 4nm-node technology

The number is still getting larger, described by “**Moore's Law**”, though slowing down

80% of 5nm chips are used in smartphones



<https://www.trendforce.com/presscenter/news/20210831-10914.html>

<https://www.tsmc.com/english/dedicatedFoundry/manufacturing/engineering>

What are the likely causes of chip shortages?

1. Skyrocketed demand due to stay-at-home orders



<https://tropico-project.eu/home-office-during-the-covid-19-pandemic-a-driver-for-digital-change>

<http://www.dualshockers.com/2014/06/04/xbox-one-june-software-update-launches-smartglass-app-for-ios-and-android-updated/>

What are the likely causes of chip shortages?

2. Surging chip demand from 5G, IoT, AI, cloud computing

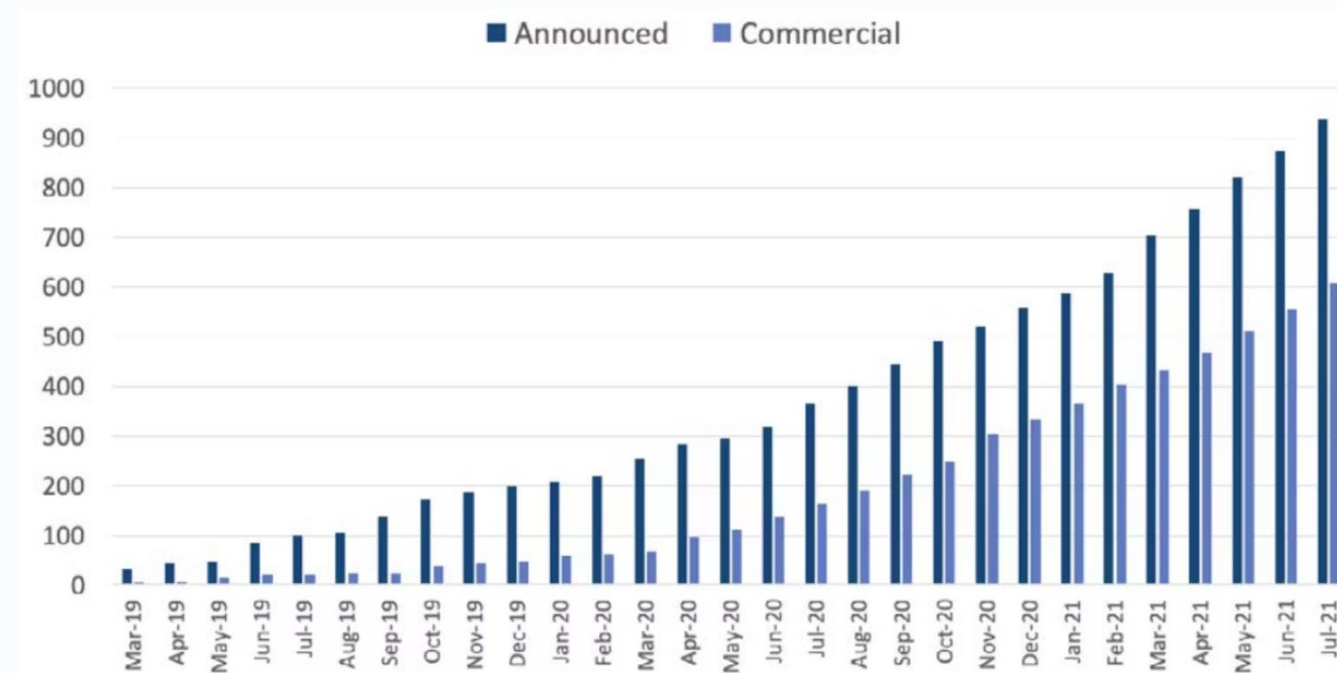
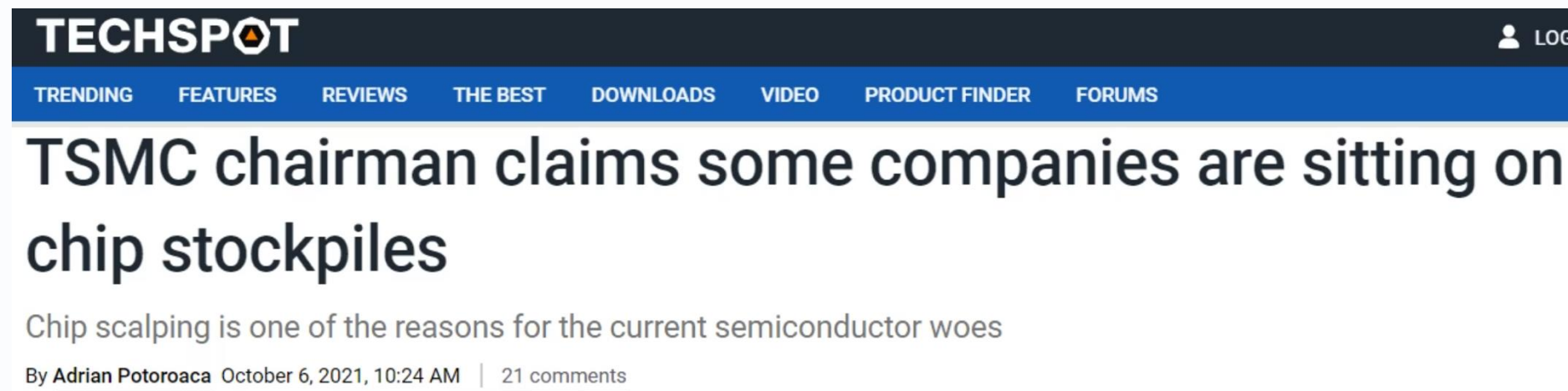


Figure 1: Growth of 5G devices
(announced and commercially available)

What are the likely causes of chip shortages?

3. Stockpiling

Some companies are stockpiling

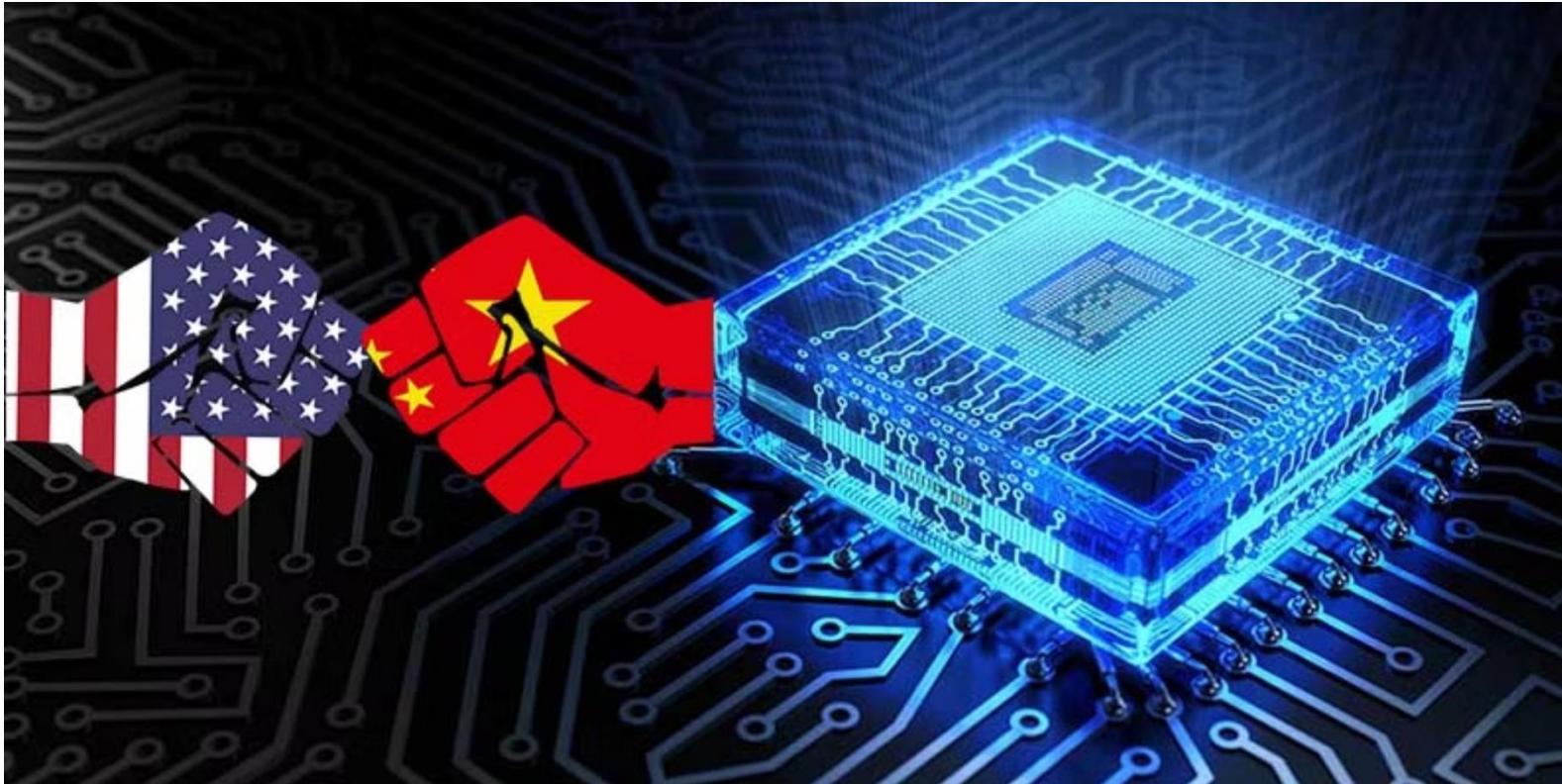


Who are they?

<https://www.techspot.com/news/91592-tsmc-chairman-claims-companies-sitting-chip-stockpiles.html><https://www.scmp.com/tech/tech-war/article/3151347/us-demand-chip-supply-data-tsmc-samsung-apple-others-sparks-debate><https://www.tomshardware.com/news/tsmc-refuses-to-disclose-clients-data-to-us-govt>

Consequences

Chips are hardware base of many new technologies important enough to be a weapon during a trade war



<https://www.go2saas.net/china-u-s-trade-war-taiwan-promised-to-support-chipmakers-industry>



Content

- Computing Evolution and Chips
- Internet and Connectivity
- Cloud Computing

Internet and the Web

What is the Internet?

- a worldwide network of interconnected computers

What is the Web?

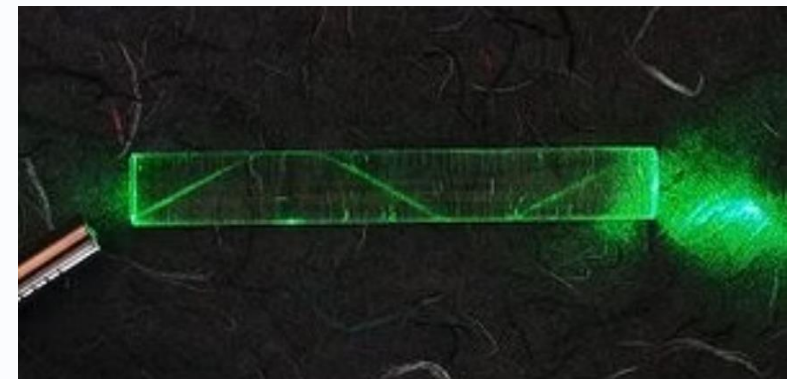
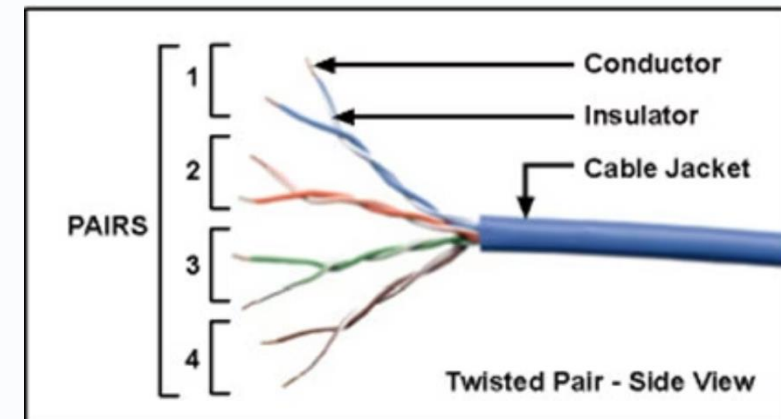
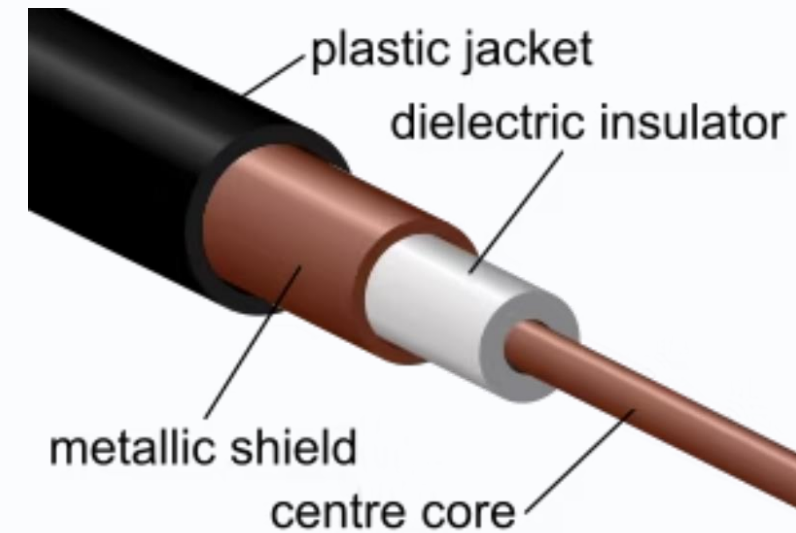
- a way of accessing information over the Internet
- The Web uses HTTP (one of the languages spoken over the Internet)
- The Web is just one of the ways to access the Internet, others: Email (SMTP), Skype (VoIP), etc.

Connectivity

Wired Communication

- Coaxial / Twisted-Pair Cable
- Optical Fiber

Invented by Charles K. Kao in 1965
(Nobel Prize in Physics, 2009) 高錕教授：
香港中文大學1987-1996校長，光纖之父
(2009年諾貝爾獎得主)

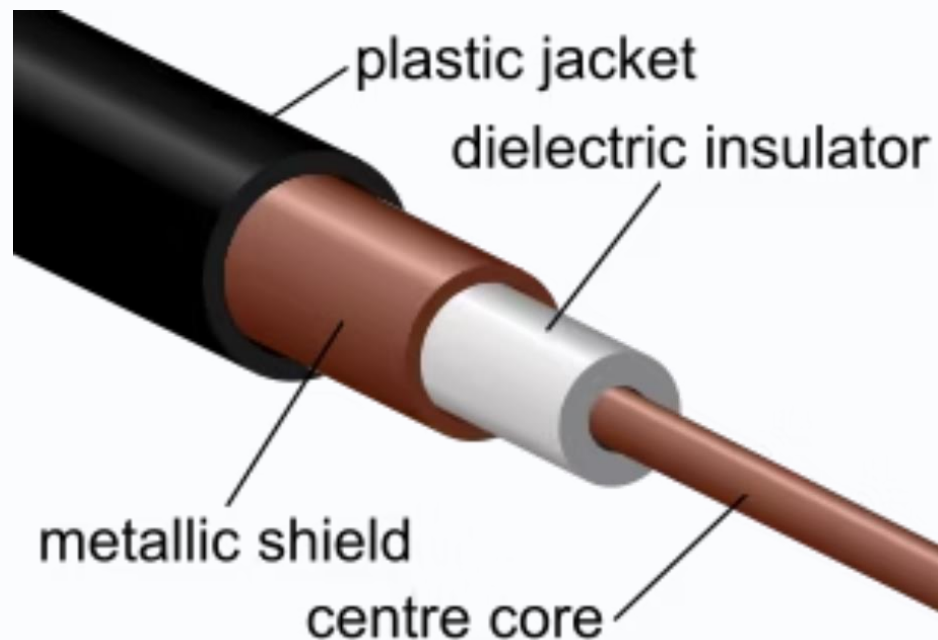


Connectivity

Wired Communication

- Coaxial: 2.3×10^8 m/s
- Optical Fiber: 2.0×10^8 m/s

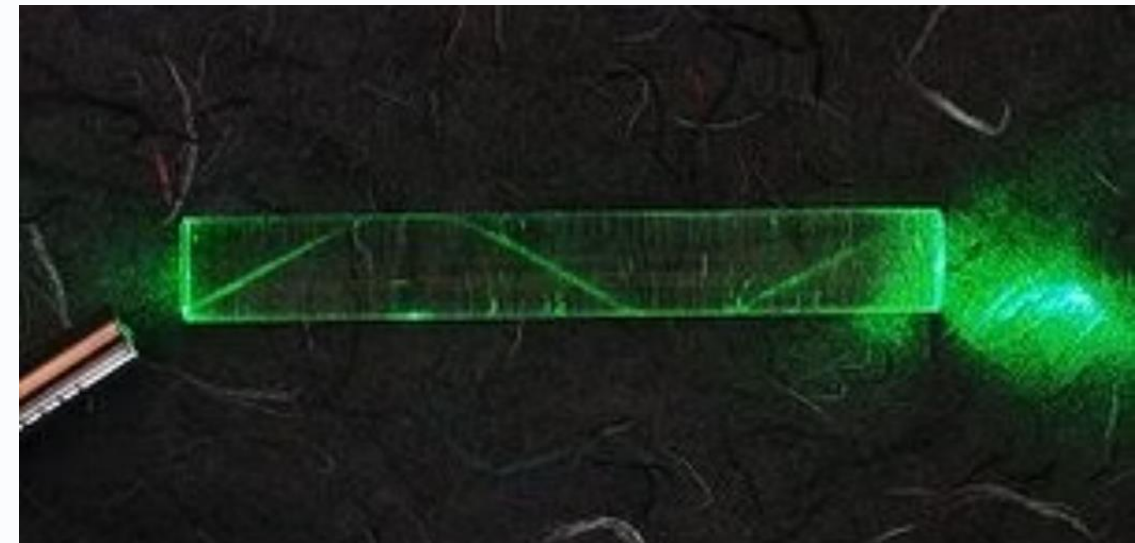
Short distance



Signal attenuation(衰减)

- Coaxial: a fast **car** in the city
- Optical Fiber: a high-speed **train**

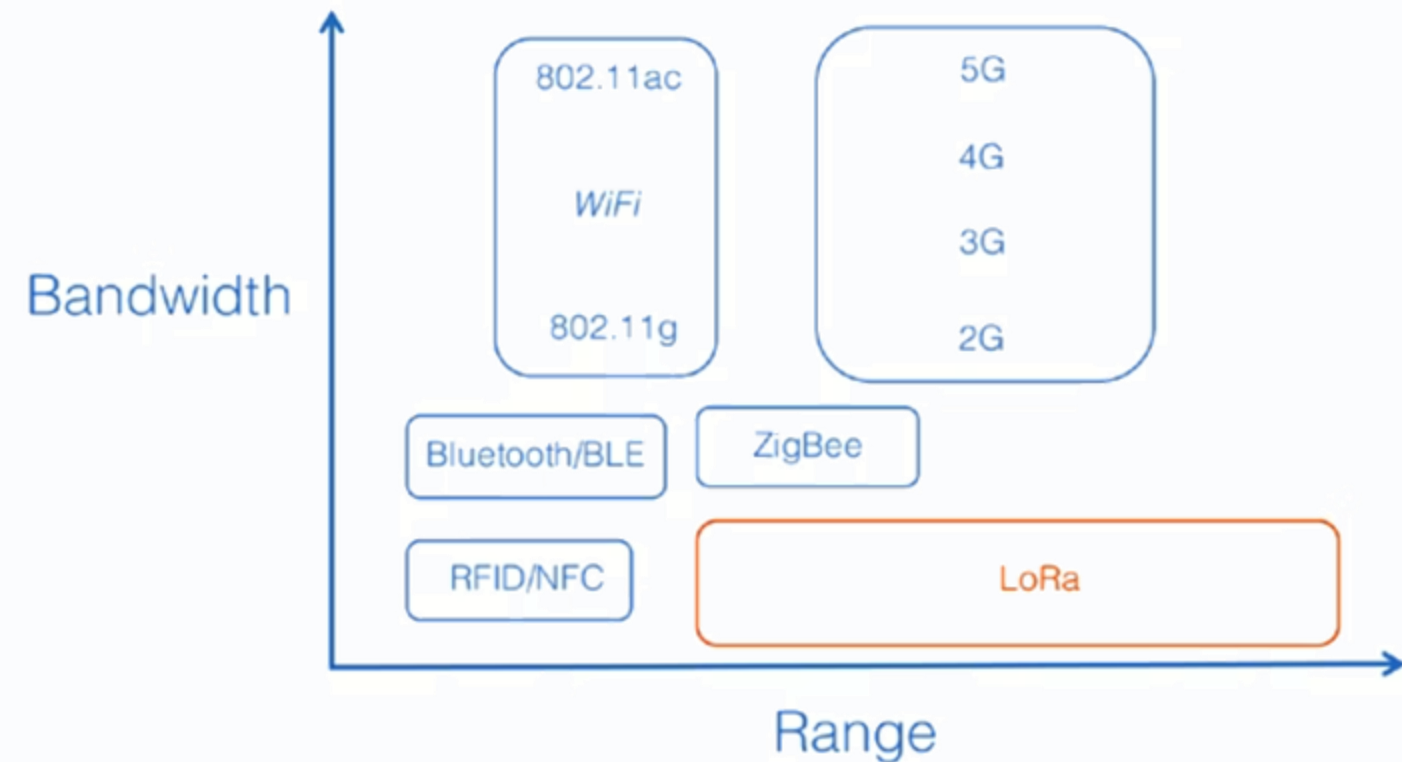
Long distance



Connectivity

Wireless Communication

- Microwave Link (e.g., Satellite)
- GSM, WCDMA, WiFi, WiMax, 3G, 4G, 5G
- Bluetooth and Near-Field Communication (NFC)



Connectivity

Pros:

- Low deployment cost
- Uses existing telephone cabling
- Will continue to work in the case of an outage

Cons:

- Subscriber speeds attenuate the farther they are from the central office
- Copper cables require maintenance and replacement after a span of time

DSL



Pros:

- Faster than DSL, Coaxial Cable, Wi-Fi
- Most future-proof medium
- Light technology makes for a noise-free networking environment

Cons:

- Very intrusive to deploy
- High deployment costs

FIBER



Pros:

- Faster than DSL
- Low deployment costs
- Uses existing coaxial cabling

Cons:

- Will not work in the case of a power outage
- Shared line architecture: Users may experience slower speeds during peak usage (but coaxial connections will usually offer higher average throughput than DSL)

COAXIAL CABLE



Pros:

- Highly mobile
- Ideal for frequent travelers
- 2-6 Mbps

Cons:

- Coverage depends on location
- Susceptible to interference

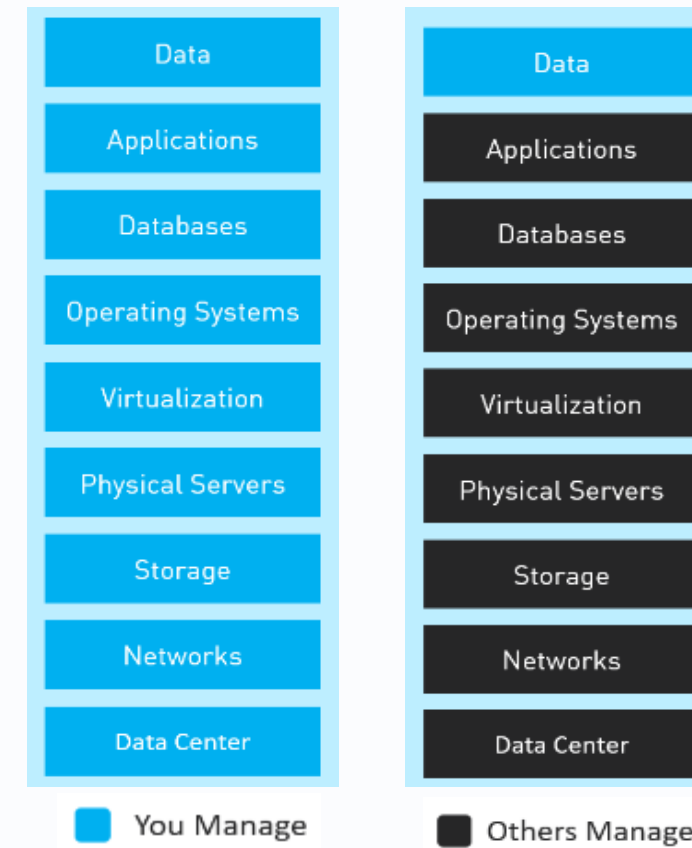
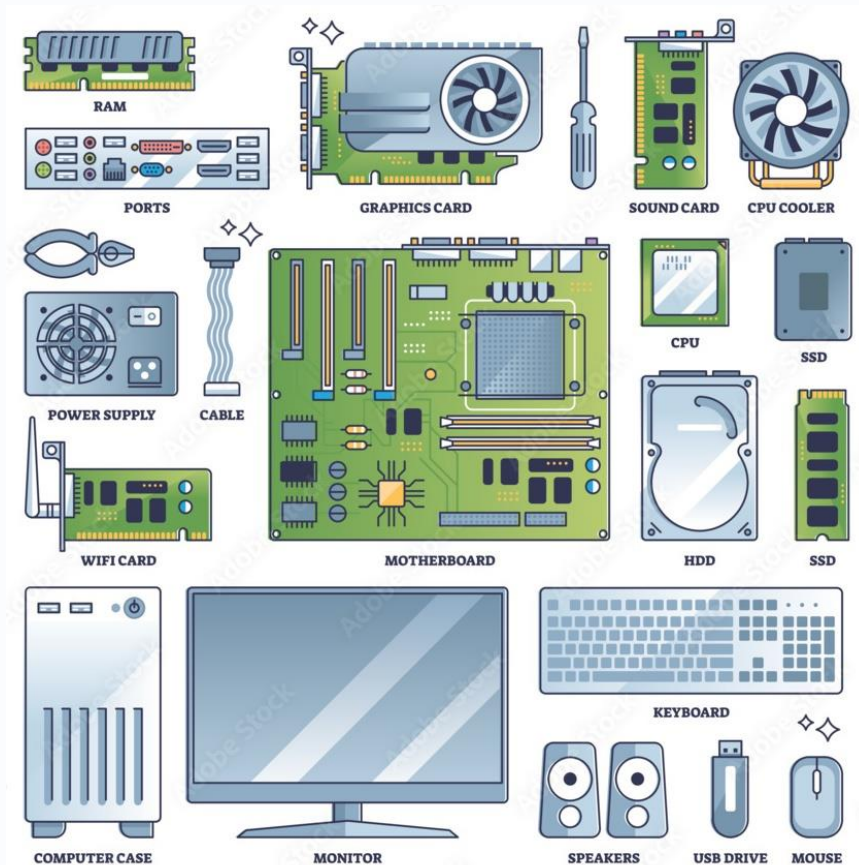
WIRELESS



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Traditional Computing



1. Fixed Hardware Specs

3. Scalability

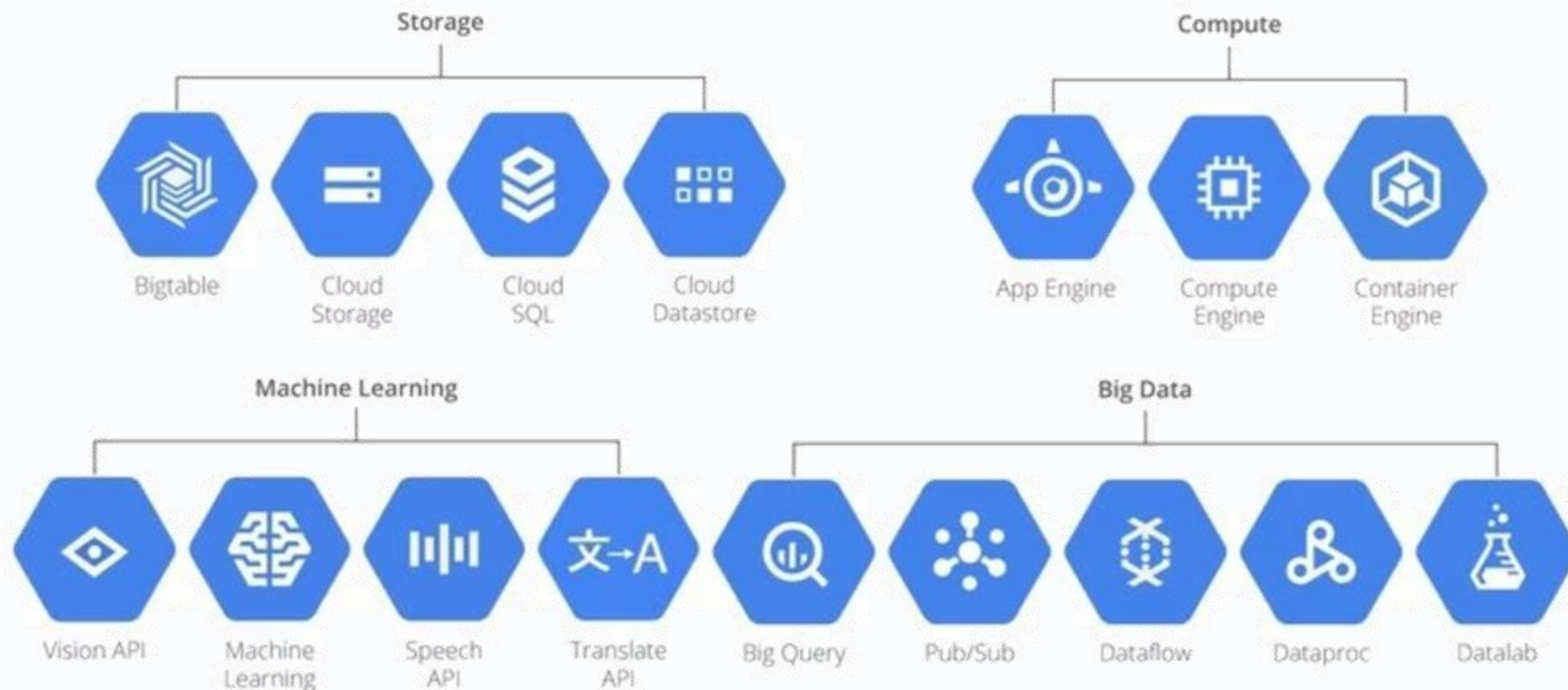
2. High upfront costs

Traditional Computing

- **Fixed hardware specs:** components such as processor, memory, hard disk, screen, ports, motherboard, etc. generally remain unchanged throughout the lifetime
- **High upfront costs:** traditional computing typically requires a significant upfront investment in hardware and software
- **Scalability:** all files, configurations, and programs are stored in one location -> locally on the computer/smartphone

Cloud Computing

- Cloud computing is an on-demand computer system
- Computing resources, such as powerful processing and storage, are usually provided by an external service provider



Traditional vs Cloud

	Architecture	Pros	Cons
Cloud	• Deliver different services via Internet on different servers	• Reduced costs • Scalability and Flexibility • Unlimited capability • Backup	• Dependence on Internet • Bandwidth • Power consumption • Privacy, security
Tradition	• Deliver different services on local server	• Reliability • Quick actions • Keeping sensitive data!	• Scalability • High upfront costs • Maintenance • Disaster recovery

Summary

- Computing Evolution and Chips
- Sensors and IoT
- Internet and Connectivity
- Big Data
- Cloud Computing
- ...



Artificial Intelligence

IT developments represent the evolution of key technologies that provide the infrastructure, tools, and context for AI to thrive

Interdependency of Technologies

AI cannot exist without IT; they are interdependent. For example:

- Data: The rise of the Internet and sensors created the "data explosion" that fuels AI
- Computation: IT advancements in processors and GPUs made it computationally feasible to train big AI models
- Connectivity: Innovations such as telecommunications and cloud computing allow AI applications to scale globally

Recommended Reading

<https://www.sciencedirect.com/topics/computer-science/chip-technology>

<https://www.bloomberg.com/explainers/global-chip-wars-semiconductors>

<https://www.asml.com/en/technology/all-about-microchips/microchip-basics>

<https://www.nytimes.com/2023/08/21/technology/nvidia-ai-chips-gpu.html>

<https://www.ibm.com/think/topics/cloud-computing>

Questions?

Contact me via: linyongqian@cuhk.edu.cn

Course Feedback

